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Foot strike angle as key predictor of multi-plane knee joint moments in sidestep cutting: implications for ACL injury prevention

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ABSTRACT

Video analyses of ACL injury situations in change-of-direction tasks suggest an elevated injury risk during flatfooted or rearfoot landing. While loading in each individual plane of the knee can contribute to injury, combined loading seems to be the most likely mechanism. This study investigated if knee joint moments are related to the foot strike angle in pre-planned and unplanned handball-specific sidestep cuts performed by $n = 51$ female handball players. Canonical correlation analysis (CCA) with the foot strike angle at initial ground contact serving as the predictor variable was performed on the knee joint moment vector field (M_{xyz}). In both conditions, the foot strike angle was related to M_{xyz}. Subsequently, post-hoc tests were conducted using CCA for moment couples (M_{xy}, M_{xz}, M_{yz}) and linear regression for individual moment components. Results revealed that traditional analyses focusing on individual planes only partially explain the observed relationships, particularly during the unplanned cutting condition. Given the time periods in which correlations exist, this study shows that manipulating the foot strike angle has the potential to reduce multi-plane knee loading associated with ACL injury, while limiting analyses to individual planes might underestimate the potential to mitigate ACL injury risk.

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ACL; injury; prevention; sidestep cutting

Introduction

Since non-contact anterior cruciate ligament (ACL) injuries occur rapidly after the initial contact (IC) of the foot with the ground (Koga et al., 2010), understanding risky kinematics at IC is key for the development of injury prevention programmes. Boden et al. (2009) and Koga et al. (2018) have suggested that injury situations typically involve the foot firmly planted on the floor. In a study with 184 female elite handball players performing pre-planned sidestep cuts, Kristianslund et al. (2014) showed that peak external knee abduction moments can be decreased by 13% when increasing the foot strike angle (FSA), i.e., the sagittal plane foot-floor angle, by one standard deviation towards a forefoot strike.

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To further understand the biomechanical factors contributing to ACL injury beyond foot–ground interactions, it is essential to consider the specific loading mechanisms at the knee joint. While anterior tibial shear force is recognised as the primary loading mechanism of the ACL, particularly at knee flexion angles below 30° (Beaulieu et al. (2023), dynamic movements such as sidestep cutting do not seem to produce sufficient sagittal plane forces to independently cause an ACL rupture (McLean et al., 2004). External knee abduction moments seem to be a key contributor in the injury mechanism (Hewett et al., 2005; Koga et al., 2010), but it has been shown that combined loads elevate ACL strain more than single-plane loading (Beaulieu et al., 2023; Markolf et al., 1995). The order of importance of knee internal rotation and abduction moments in contributing to ACL injury is a topic of debate in the literature (Beaulieu et al., 2023). While it has been demonstrated that these loads significantly strain the ACL both in isolation and combination (Markolf et al., 1995), either of these loads could potentially exceed ACL injury thresholds during sidestep cutting (McLean et al., 2004). Nevertheless, the combination of these loads seems to be especially prone to induce high stress on the ACL and is most likely the most dangerous (Oh et al., 2012; Shin et al., 2011). Hence, analysing individual components or planes might be a reductionist approach to the three-dimensional loading situation occurring during real injury situations like dynamic change-of-direction tasks, including sidestep cutting, as these show a combination of knee valgus, tibia internal rotation and anterior translation (Koga et al., 2010). However, research investigating the relationship between FSA and the multi-plane knee moment vectors is sparse.

Building on the need for a multi-planar perspective, Robinson et al. (2015) have previously demonstrated how correlations between knee flexion angle and individual moment components do not account for the complexity of the three-dimensional knee moment in an unplanned sidestep cutting manoeuvre performed by either females or males. Furthermore, isolating the analysis to knee abduction moments might be less sensitive to identifying at-risk athletes during sportive movements like landings and pre-planned sidestepping compared to analyses of non-sagittal moment vectors (Robinson et al., 2023). Extending this line of thought to preventive measures, assessing the effect of technique used to perform sportive manoeuvres might underestimate the relationship with injury-prone loading when moment components are investigated individually.

These findings indicate that vector field analyses transcend traditional analyses by offering a more comprehensive understanding of the relationship between movement technique and joint loading. Therefore, to better understand how FSA affects knee moments associated with ACL injury, this study explored the relationship between FSA and the knee moment vector field during both pre-planned and unplanned sidestep cuts. Since pre-planned and unplanned cutting manoeuvres differ in terms of muscle activation and movement technique (Besier et al., 2003; Bill et al., 2022; Meinerz et al., 2015), examining both conditions independently may allow for generalisability of the observed relationship across different types of movement conditions. Given the disproportionately higher incidence of ACL injuries in female athletes (Smith et al., 2012), the study specifically focused on exploring this relationship within a female cohort. We hypothesised that there is no correlation between FSA and knee moments. Given the non-directional nature of this hypothesis—without specification to a particular plane or

time point—we tested it by analysing knee moment vector fields across the entire stance phase (Pataky et al., 2013).

Materials and methods

Participants

Data were collected from $n = 51$ high-level female handball players competing at levels ranging from Elite to the 3rd Division (mean ± SD: 67.0 ± 7.7 kg, 1.70 ± 0.06 m, 19.2 ± 3.4 years). The sample size was not based on an *a priori* power analysis but was determined based on resource constraints (Lakens, 2022). At the time of testing, all players were at least 16 years of age and injury- and pain-free. The study was approved by the University Ethics Committee in compliance with the Declaration of Helsinki. Written informed consent to participate in this study was provided by all players or their legal guardian.

Design and protocol

As part of a larger study (Mai et al., 2022), players were equipped with a full-body marker set, and kinematics and kinetics were sampled with a 24-camera motion capture system (Qualisys, Gothenburg, Sweden; 200 hz) and two floor-embedded force platforms (AMTI, Watertown, Massachusetts, USA, 600 × 1200 mm; 1000 hz), respectively. Players performed a standardised warm-up protocol followed by a familiarisation period. Subsequently, a minimum of five valid trials, i.e., following a clean catch of the ball with the foot fully placed within the boundaries of the force platform, of each of the following two conditions were collected: 1) a pre-planned handball-specific cutting manoeuvre (PP), in which they were instructed to catch a ball—passed by a teammate—during the penultimate step and subsequently execute a sidestep cut to feint and pass a stationary human defender (Figure 1(a)), and 2) an unplanned handball-specific cutting manoeuvre (UP), in which they were instructed to catch a ball—passed by a teammate—during the penultimate step and subsequently perform a sidestep cut to feint and pass three variably moving defenders (Figure 1(b)). In this task, the middle defender and one of the outside defenders pressured the player upon catching the ball, and the player was instructed to cut towards the side of the defender that remained static, resulting in an unplanned cutting direction (Mai et al., 2022). However, only cuts performed with the same leg as during PP, i.e., the leg ipsilateral to the throwing arm, were included in the analysis. Cutting angles and approach speeds were not controlled, but all players approached the force platform from a standardised starting position with a six-metre run-up at an approximately 35° angle to its long axis at match-like intensities. This resulted in mean cutting angles of 69.2 ± 14.9° and 61.5 ± 14.0°, and approach speeds of 3.15 ± 0.14 m/s and 2.96 ± 0.35 m/s for the pre-planned (PP) and unplanned (UP) conditions, respectively, as determined retrospectively from the horizontal centre of mass velocity vectors after data collection.

Data analysis

IC and toe-off (TO) were defined as the time points when the unfiltered vertical ground reaction force (GRF) component exceeded or fell below 30 N, respectively. A recursive

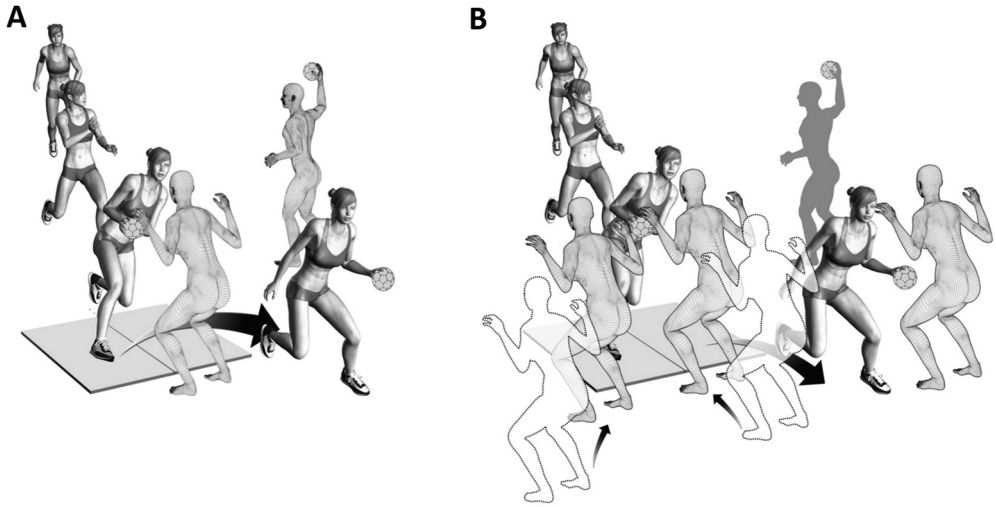

Figure 1. Illustrations of the sidestep cutting manoeuvres. (a) Preplanned cut (PP). (b) Unplanned cut (UP). Reprinted from Bill et al. (2022), licenced under CC BY.

4th-order low-pass Butterworth filter (20 hz cut-off frequency) was applied to the raw marker trajectories and GRF (Kristianslund et al., 2012; Mai & Willwacher, 2019). Ankle and knee joint centres were defined as the midpoints between medial and lateral malleoli and femoral condyle markers, respectively. Hip joint centres and coordinate systems were defined according to Bell et al. (1990) and Seidel et al. (1995). Segment inertial properties were calculated based on anthropometric data derived from (de Leva, 1996). Lower extremity resultant external joint moments were determined with the explicit expression provided by Hof (1992) using a rigid body model of the lower extremities. All model calculations were performed using a custom-made MATLAB script (R2021b, The Mathworks, Natick, USA). Details of these calculations and the marker set used can be found in previous publications (Willwacher et al., 2014, 2016).

For each player and task, an overall resultant external knee joint moment vector field (M_{xyz}) was created after time-registering all data to 201 data points from IC to TO, averaging five trials, and normalising to body mass. The whole stance phase was considered for analysis as the hypothesis did not pertain to a specific time window. The components of the vector field were defined as follows: M_x representing frontal plane moments, M_y representing sagittal plane moments, and M_z representing transverse plane moments (see Figures 2(a) and 3(a)). In addition, non-transverse (M_{xy}), non-sagittal (M_{xz}), and non-frontal (M_{yz}) plane knee moment vector fields were also generated. FSA was defined as the angle between the long axis of the foot and its projection to the ground at IC, with positive angles indicating a forefoot strike (see Figures 2(a) and 3(a)).

Statistical analysis

To test the null hypothesis that FSA and knee joint moments are unrelated, canonical correlation analysis (CCA), the statistical parametric mapping (SPM) vector field equivalent of linear regression (Pataky et al., 2013), was performed with FSA serving as the

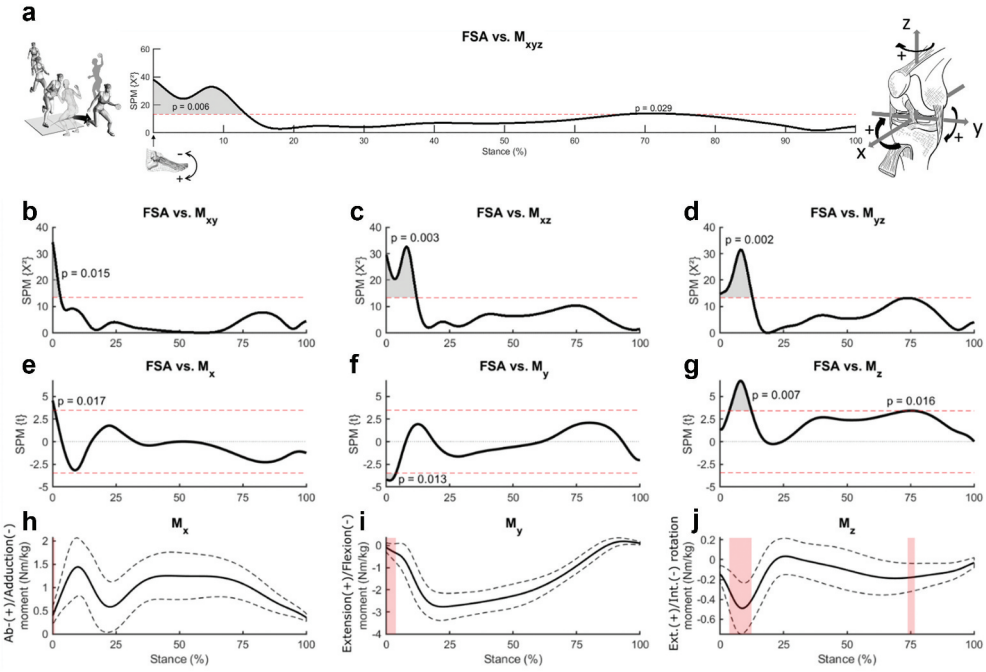

Figure 2. Results for the pre-planned (PP) cutting manoeuvre. a) illustration of the task (left), trajectory level SPM analyses for the relationship between the foot strike angle and the overall M_{xyz} vector field (centre), and knee coordinate system definition (right). B—D) trajectory level SPM analyses for the relationship between the foot strike angle and the moment couples M_{xy}, M_{xz}, and M_{yz}. E—G) trajectory level SPM analyses, depicting the strength of positive (+) and negative (-) correlation between the individual moment components (M_x, M_y, and M_z) and the foot strike angle. H—J): resultant external knee joint moments M_x, M_y, and M_z. Light red-shaded areas indicate time intervals with significant correlations.

predictor variable. The alpha level was set to $\alpha = 0.05$. CCA was also performed for post-hoc analyses of moment couples (M_{xy}, M_{xz}, M_{yz}), and linear regression was performed as post-hoc tests for individual moment components (M_x, M_y, M_z). To retain a family-wise error rate of $\alpha = 0.05$, Šidák thresholds of 0.017 were adopted for each post-hoc test level (Šidák, 1967). Statistics were calculated using statistical parametric mapping (SPM, v. M0.4.8, www.spm1d.org) for MATLAB (R2021b, The Mathworks, Natick, USA).

Results

Mean FSA for PP and UP were −2.1 ± 13.5° and 6.8 ± 9.9°, respectively, and ranged from −27.8° to 21.0° and −15.4° to 20.2°, respectively.

For PP, FSA was significantly related to M_{xyz} during early ($p = 0.006$ at 0–13.2%) and late stance ($p = 0.029$ at 67.7–74.6%; Figure 2(A)). Post-hoc analyses of moment couples revealed that the non-sagittal (M_{xz}; $p = 0.003$ at 0–12.0%; Figure 2(C)) and non-frontal (M_{yz}; $p = 0.002$ at 0–12.8%; Figure 2(D)) moment couples explained most of the results found during early stance while results for the non-transverse (M_{xy}) moment couple were also significant ($p = 0.015$ at 0–3.1%; Figure 2(B)).

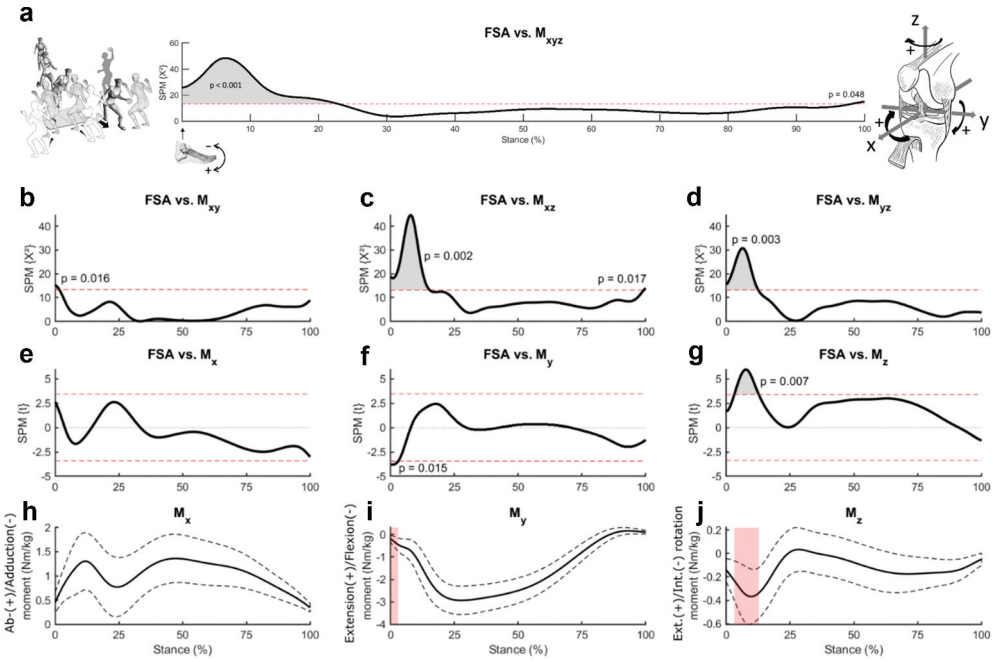

Figure 3. Results for the unplanned cutting manoeuvre. A) illustration of the task (left), trajectory level SPM analyses for the relationship between the foot strike angle and the overall M_{xyz} vector field (centre), and knee coordinate system definition (right). B—D) trajectory level SPM analyses for the relationship between the foot strike angle and the moment couples M_{xy}, M_{xz}, and M_{yz}. E—G) trajectory level SPM analyses, depicting the strength of positive (+) and negative (-) correlation between the individual moment components (M_x, M_y, and M_z) and the foot strike angle. H—J): resultant external knee joint moments M_x, M_y, and M_z. Light red-shaded areas indicate time intervals with significant correlations.

Individually, the transverse plane moment (M_z; Figure 2(j)) explained most of the results found during early ($p = 0.007$ at 3.6–12.4%) and late stance ($p = 0.016$ at 73.8–6.6%, Figure 2(g)). Results for the frontal (M_x; Figure 2(h)) and sagittal plane moment (M_y; Figure 2(I)) also yielded significance during a brief early portion of stance ($p = 0.017$ at 0–0.8%; Figure 2(e) and $p = 0.013$ at 0–3.8%; Figure 2(f), respectively).

For UP, FSA was significantly related to M_{xyz} during early ($p < 0.001$ at 0–22.9%) and late stance ($p = 0.048$ at 98.0–100%; Figure 3(a)).

Post-hoc analyses of moment couples revealed that the non-sagittal (M_{xz}; $p = 0.002$ at 0–15.3%; Figure 3(c)) and sagittal-transverse (M_{yz}) moment couples ($p = 0.003$ at 0–12.5%; Figure 3(d)) partially explained the results found during early stance while results for the frontal-sagittal (M_{xy}) moment couple were also significant ($p = 0.016$ at 0–1.6%; Figure 3(b)). The non-sagittal moment couple (M_{xz}) partially explained the results during push-off ($p = 0.017$ at 99.3–100%; Figure 3(c)).

Individually, it was mainly the transverse plane moment (M_z; Figure 3(j)) that explained the results found during early stance ($p = 0.007$ at 3.2–12.8%, Figure 3(g)) while results for the sagittal plane moment (M_y; Figure 3(i)) also yielded significance ($p =$

0.015 at 0–2.9%; Figure 3(f)). However, moment couples and individual moment components failed to provide a complete explanation of the results.

Discussion and implications

The purpose of this study was to investigate the relationship between FSA and the overall M_{xyz} moment vector field throughout stance during both pre-planned and unplanned sidestep cuts. The study revealed that FSA is significantly correlated with the M_{xyz} vector field in PP and UP. Correlations were found during both early and late stance, with the latter correlations most likely being irrelevant in the context of non-contact ACL injuries (Koga et al., 2010). Nevertheless, we analysed the entire stance phase, as our hypothesis was not limited to a specific anatomical plane or time point. This comprehensive approach was necessary to avoid inter-component covariance bias and regional focus bias (Pataky et al., 2013).

In both cutting scenarios, it was the non-sagittal and the non-frontal plane moment couples, M_{xz} and M_{yz}, respectively, that best explained the results found for the three-component vector, M_{xyz}. This effect may either arise because forefoot striking alters the magnitude of the GRF (Bill et al., 2024), likely due to energy absorption through the Achilles tendon-triceps surae muscle-tendon unit, or the moment arm (Kristianslund et al., 2014) of the GRF vector (or both). Individually, the transverse plane moment, M_z, showed the highest correlation with FSA. However, moment couples and individual moment components failed to fully explain the correlations found for the overall M_{xyz} vector fields. This discrepancy can be attributed to the complex three-dimensional nature of knee moments, which is captured in the M_{xyz} vector field analysis but overlooked when examining individual components.

Given that the non-sagittal plane moments are highly relevant for ACL injury (Koga et al., 2010; McLean et al., 2004), and FSA is related to multi-plane knee moments with a particularly strong correlation in the non-sagittal plane combination, FSA may have the potential to modify ACL injury risk. These insights enhance our understanding of the biomechanical implications of initial ground contact kinematics on knee loading patterns, shedding light on factors influencing ACL injury risk.

Previous research has highlighted the role of the knee abduction moments (Hewett et al., 2009; Koga et al., 2010; Krosshaug et al., 2007) and flatfooted landing (Boden et al., 2009; Koga et al., 2018) in contributing to the ACL injury mechanism. On an individual moment component level, however, FSA showed the strongest correlation with transverse plane moments in the present study. As the knee internal rotation moment substantially strains the ACL (Markolf et al., 1995; Oh et al., 2012), focusing solely on frontal plane moments could be a reductionist approach. Ignoring key interactions between planes might therefore not only neglect critical factors contributing to ACL injury risk, but also underestimate the effect of technique adaptations on ACL injury-relevant knee joint loading. Hence, vector field analyses offer a more comprehensive understanding of the relationship between movement technique and joint loading than traditional analyses that focus on individual planes.

Our findings regarding the moment couples align with the recent results of Ogasawara et al. (2024), who demonstrated that when striking the ground with the forefoot, the GRF vector is less likely to induce a combined knee abduction-internal rotation moment. In

contrast, the authors reported that heel-first landings show a significantly higher occurrence rate of the non-sagittal plane loading combination due to the GRF's vector direction. Our results reveal strong correlations in the non-sagittal moment couple, particularly in the UP condition. In unplanned conditions, where movements are typically more variable (Weir et al., 2019), the body's ability to adjust dynamically could amplify the protective effects of forefoot striking. On an individual moment component level, an increase in FSA (i.e., towards a forefoot strike) seems to be inversely related to both knee abduction moments (albeit non-significantly; Figure 2(e,h), Figure 3 (e,h)) and internal rotation moments (significantly; Figure 2(g,j) Figure 3(g,j)) in the phase of peak loading. Considering evidence from Quatman et al. (2010) that knee abduction and internal rotation moments often occur together in athletic manoeuvres, and studies by Markolf et al. (1995) and Oh et al. (2012) showing that these combined moments strain the ACL substantially more than when they occur in isolation, our results provide additional support for forefoot striking as a mechanism to reduce ACL injury-relevant knee joint loads.

Recent research investigating the effect of FSA modification on knee abduction moments showed promising results (Bill et al., 2024). However, some of the players that were investigated were unsuccessful in reducing their knee abduction moments despite increasing their FSA towards a (more pronounced) forefoot strike. Results of the present study indicate that improvements might go beyond reductions in single-plane loads, and research investigating moment couples or three-dimensional knee moment vectors might be better suitable to assess the effects of technique modification on loads associated with ACL injury as single-plane analyses might underestimate improvements in the context of injury risk. However, while individual joint moments can be easily interpreted in terms of potential injury risk, translating three-dimensional moment vectors into clear movement-based recommendations is more challenging. This complexity underscores that decomposing the three-dimensional moment into its dominant contributors may improve practical interpretation. Therefore, communicating movement recommendations based on individual plane analyses remains valuable, provided that coaches and practitioners recognise the limitations of this reductionist approach and acknowledge that single-plane assessments may underestimate the multi-planar nature of injury risk. Furthermore, highlighting the potential performance-enhancing benefits of forefoot striking may make movement modifications more appealing to athletes and coaches, thereby increasing the likelihood of successful implementation (Fox, 2018).

Moment couples (mainly during UP) and individual moment components (during PP and UP) failed to fully explain the correlations between FSA and M_{xyz}, as they do not account for covariance or capture the full temporal effects of the vector. This limitation may increase the risk of Type II errors (Pataky et al., 2013). Another disadvantage of analysing individual planes is their relatively high waveform variance during sidestep cuts, especially in the frontal plane (Sankey et al., 2015), and their sensitivity to the anatomical coordinate system definition (Robinson et al., 2023). Vector field results, on the other hand, suggest that the coordinate system into which the moment vector is resolved is unrelated to the moment direction of interest (Robinson et al., 2015).

This study is not without limitations. First, while the evidence strongly supports the positive effects of forefoot striking on ACL-injury-relevant knee loads, no direct injury data were collected. Additionally, the impact on the ankle joint and its associated injury

risk falls outside the scope of this research. Future prospective studies are required to establish a causal relationship with injury risk. Second, while results for PP and UP were generally very similar, it is unclear why the discrepancies between M_{xyz} and moment couples and components were higher in UP than PP and what factors contributed. Last, this study was solely focused on female handball players. Therefore, it remains unclear if the results can be transferred to male populations or athletes from other sports.

In conclusion, FSA is a predictor for three-dimensional knee moments in both PP and UP. While moment couples and individual components provide some understanding about the relationship with FSA, the overall M_{xyz} vector field offers a broader perspective, particularly in the UP scenario. The correlations identified during the early peak loading phase indicate that employing a forefoot strike might be beneficial in mitigating ACL injury-relevant joint loads. Therefore, incorporating forefoot strike techniques into injury prevention programmes could be a strategic approach to reducing ACL injury risk.

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